Task 1: Enable MFA on the AWS account

Procedure:
1. Sign in to the AWS Management Console.
2. Open IAM → Users and select your own IAM user, or open the account's security credentials if you use the root account.
3. Find the Multi-factor authentication (MFA) section.
4. Select Assign MFA device.
5. Choose an authenticator app or other supported virtual MFA device.
6. Follow the prompts to scan the QR code and enter the required authentication codes.
7. Complete the setup and verify that the MFA device is assigned.

Task 2: Test a new browser session

Procedure:
1. Open a private/incognito window or a different browser.
2. Navigate to the AWS Management Console sign-in page.
3. Sign in with the relevant account credentials.
4. If prompted, enter the code from the configured MFA device.
5. Record the prompts and whether the sign-in succeeds.

Observation: When MFA is required for the identity and sign-in flow, an additional verification step is presented. Exact prompts depend on the account type and authentication configuration.

Task 3: Compare access with and without MFA

MFA enabled	MFA disabled
Requires a second factor when the sign-in flow calls for it.	Does not require that MFA factor for sign-in.
Adds protection if a password is stolen.	A stolen password may be sufficient if no other control blocks access.

Helps reduce account takeover risk.	Provides less protection against credential theft.

Task 4: Attacker scenario

An attacker may obtain a cloud user's password through phishing, password reuse, or a data breach. The attacker then attempts to sign in to the cloud account. If MFA is enabled and required, the stolen password alone is generally insufficient to complete authentication. Suspicious sign-in activity may also trigger alerts if monitoring and detection rules are configured.